

Laplace transzformáció

① $f_1(t) = \frac{d}{dt} \cos(\omega t) \varepsilon(t)$

a) deriválás $\rightarrow$ Laplace

$$f_1(t) = -\omega \cdot \sin(\omega t) \xrightarrow{\mathcal{L}} F_1(s) = -\omega \underbrace{\mathcal{L}\{\sin(\omega t)\}}_{\frac{\omega}{s^2 + \omega^2}} = -\frac{\omega^2}{s^2 + \omega^2}$$

b) deriválási szabály

$$\mathcal{L}\left\{\frac{d}{dt} f(t)\right\} = sF(s) - f(0)$$

$$F(s) = \mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$f(0) = \cos(\omega \cdot 0) = \cos(0) = 1$$

$$F_1(s) = \frac{s^2}{s^2 + \omega^2} - 1 = \frac{s^2}{s^2 + \omega^2} - \frac{s^2 + \omega^2}{s^2 + \omega^2} = -\frac{\omega^2}{s^2 + \omega^2}$$

② $f_2(t) = \frac{d}{dt} \sin(\omega t + \varphi) \varepsilon(t)$

szinusz tétel

$$\sin(\alpha + \beta) = \sin\alpha \cdot \cos\beta + \sin\beta \cdot \cos\alpha$$

$$\sin(\omega t + \varphi) = \sin(\omega t) \cos\varphi + \sin\varphi \cos(\omega t)$$

a) deriválás

$$f_2(t) = (\cos\varphi \cdot \omega \cdot \cos(\omega t) - \sin\varphi \cdot \omega \cdot \sin(\omega t)) \varepsilon(t)$$

$\downarrow \mathcal{L}$

$$F_2(s) = \cos\varphi \cdot \omega \cdot \frac{s}{s^2 + \omega^2} - \sin\varphi \cdot \omega \cdot \frac{\omega}{s^2 + \omega^2} = \frac{\omega}{s^2 + \omega^2} (\cos\varphi \cdot s - \sin\varphi \cdot \omega)$$

b) deriválási szabály

$$F_2(s) = sF(s) - f(0)$$

$$F(s) = \cos\varphi \frac{\omega}{s^2 + \omega^2} + \sin\varphi \frac{s}{s^2 + \omega^2}$$

$$f(0) = \cos\varphi \underbrace{\sin(0)}_0 + \sin\varphi \underbrace{\cos(0)}_1 = \sin\varphi$$

$$F_2(s) = \underbrace{s \cdot \cos\varphi \frac{\omega}{s^2 + \omega^2} + s \cdot \sin\varphi \frac{s}{s^2 + \omega^2}}_{sF(s)} - \underbrace{\sin\varphi}_{f(0)}$$

$$= \cos \varphi \frac{sw}{s^2+w^2} + \sin \varphi \frac{s^2}{s^2+w^2} - \sin \varphi \cdot \frac{s^2+w^2}{s^2+w^2} -$$

$$\sin \varphi \cdot \frac{s^2 - s^2 - w^2}{s^2+w^2} = - \sin \varphi \frac{w^2}{s^2+w^2}$$

$$= \frac{w}{s^2+w^2} (\cos \varphi \cdot s - \sin \varphi \cdot w)$$

$$(3) f_3(t) = 12 \int_0^t e^{2\tau+1} \cos(3\tau) d\tau$$

integrációs szabály!

$$\mathcal{L} \left\{ \int_0^t f(\tau) d\tau \right\} = \frac{1}{s} F(s)$$

$$F(s) = \mathcal{L} \left\{ \underbrace{e^{2\tau+1}}_{e^{2\tau} \cdot e} \cos(3\tau) \right\} = e \cdot \mathcal{L} \left\{ \underline{e^{2\tau}} \cdot \cos(3\tau) \right\} =$$

cselekvítési szabály

$$p = -2$$

$$f(\tau) = \cos(3\tau)$$

$$\mathcal{L} \{ f(t) \cdot e^{-pt} \} = F(s) \Big|_{s=s+p}$$

$$= e \left(\frac{s}{s^2+w^2} \Big|_{s=s-2} \right) = e \cdot \frac{s-2}{(s-2)^2+3^2}$$

$$w=3$$

$$F_3(s) = 12 \cdot \frac{1}{s} \cdot e \cdot \frac{s-2}{(s-2)^2+3^2} = \frac{12e}{s} \cdot \frac{s-2}{s^2-4s+13}$$

$$(4) f_4(t) = (-t^3 + 2t^2 \cdot e^t) \varepsilon(t)$$

lin. szabály

$$\mathcal{L} \{ f_4(t) \} = - \mathcal{L} \{ t^3 \} + 2 \mathcal{L} \{ t^2 \cdot e^t \}$$

$$\mathcal{L} \{ t^n \cdot \varepsilon(t) \} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L} \{ t^3 \} = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$$

általános esetben

$$\mathcal{L} \{ (-1)^n t^n f(t) \} = \frac{d^n}{ds^n} F(s)$$

valószínű való szorzás

1.) cselekvítési szabály

$$f(t) = t^2 \rightarrow F(s) = \frac{2}{s^3}$$

$$p = -1$$

$$\mathcal{L} \{ t^2 \cdot e^t \} = \frac{2}{s^3} \Big|_{s=s-1} = \frac{2}{(s-1)^3}$$

$$2., n=2-re: \mathcal{L}\left\{\underbrace{(-1)^2}_{1} t^2 f(t)\right\} = \frac{\alpha^2}{\alpha s^2} F(s)$$

$$f(t) = e^t \rightarrow F(s) = \frac{1}{s} \Big|_{s=s-1} = \frac{1}{s-1}$$

$$\mathcal{L}\{t^2 \cdot e^t\} = \left(\frac{\alpha}{\alpha s}\right) \left(\frac{\alpha}{\alpha s}\right) \frac{1}{s-1}$$

$$= \left(\frac{\alpha}{\alpha s}\right) (-1) (s-1)^{-2} = (-1) \underbrace{(-2)}_2 (s-1)^{-3} =$$

$$= \frac{2}{(s-1)^3}$$

$$F_4(s) = -\frac{6}{s^4} + \frac{4}{(s-1)^3}$$

Inverz Laplace transzformáció

$$\textcircled{1} F_1(s) = 3 \cdot \frac{1 \cdot s + 2}{(s+1)(s+4)}$$

→ parciális törtekre bontás

$$F_1(s) = 3 \left[\frac{A}{s+1} + \frac{B}{s+4} \right] = 3 \left[\frac{A(s+4) + B(s+1)}{(s+1)(s+4)} \right] = 3 \left[\frac{As + 4A + Bs + B}{(s+1)(s+4)} \right]$$

$$s^0: 4A + B = 2 \rightarrow B = 2 - 4A$$

$$s^1: A + B = 1$$

$$\downarrow$$

$$\rightarrow A + 2 - 4A = 1$$

$$1 = 3A$$

$$A = \underline{\underline{1/3}}$$

↓

$$B = 2 - 4/3 = \underline{\underline{2/3}}$$

$$F_1(s) = 3 \left[\frac{1}{3} \cdot \frac{1}{s+1} + \frac{2}{3} \frac{1}{s+4} \right] = \frac{1}{s+1} + \frac{2}{s+4}$$

↓ $\mathcal{L}^{-1}$

$$f_1(t) = \mathcal{L}^{-1}\{F(s)\} = \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}}_{\frac{1}{s} \Big|_{s=s+1}} + 2 \cdot \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\}}_{\frac{1}{s} \Big|_{s=s+4}} =$$

$$= \underline{\underline{(e^{-t} + 2e^{-4t}) \varepsilon(t)}}$$

$$(2) F_2(s) = 3 \frac{(s+2)^2}{(s+1)(s+4)} = 3 \frac{s^2 + 4s + 4}{s^2 + 5s + 4}$$

→ ha a nevező és a számláló fokrátalma azonos ($n=r$)
polinomosztás!

$$F_2(s) = 3 \frac{s^2 + 5s + 4 - s}{s^2 + 5s + 4} = 3 \left[1 - \frac{s}{s^2 + 5s + 4} \right]$$

$$\text{VAGY } (s^2 + 4s + 4) : (s^2 + 5s + 4) = 1$$

$0 - s + 0$

$$\oplus \text{ maradék: } \frac{-s}{s^2 + 5s + 4}$$

$$\frac{s^2 + 4s + 4}{s^2 + 5s + 4} = 1 - \frac{s}{s^2 + 5s + 4}$$

$$f_2(t) = \mathcal{L}^{-1}\{F_2(s)\} = 3\mathcal{L}^{-1}\{1\} - 3\mathcal{L}^{-1}\left\{\frac{1 \cdot s + 0}{(s+1)(s+4)}\right\}$$

parciális törtekre bontás

$$\mathcal{L}^{-1}\{1\} = \delta(t)$$

$$\frac{A}{s+1} + \frac{B}{s+4} = \frac{As + 4A + Bs + B}{(s+1)(s+4)}$$

$$s^0: 4A + B = 0 \rightarrow B = -4A$$

$$s^1: A + B = 1$$

$$\downarrow$$

$$A - 4A = 1$$

$$-3A = 1$$

$$A = -1/3$$

$$B = \underline{\underline{4/3}}$$

$$\mathcal{L}^{-1}\left\{-\frac{1}{3} \frac{1}{s+1} + \frac{4}{3} \frac{1}{s+4}\right\} = \frac{1}{3} \left(-\underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}}_{\frac{1}{s} \Big|_{s=s+1}} + 4 \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\}}_{\frac{1}{s} \Big|_{s=s+4}} \right)$$

$$= \frac{1}{3} (-e^{-t} + 4e^{-4t}) \varepsilon(t)$$

$$f_2(t) = 3\delta(t) + (e^{-t} - 4e^{-4t}) \varepsilon(t)$$

$$(3) F_3(s) = \frac{1-e^{-s}}{s+7} = \frac{1}{s+7} - \frac{e^{-s}}{s+7}$$

$$f_3(t) = \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+7}\right\}}_{\frac{1}{s} \Big|_{s=s+7}} - \mathcal{L}^{-1}\left\{e^{-s} \cdot \frac{1}{s+7}\right\}$$

$$\frac{1}{s} \Big|_{s=s+7}$$

$$= e^{-7t} \varepsilon(t)$$

eltolási szabály

$$\tau=1$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s+7}\right\} = e^{-7t} \varepsilon(t)$$

de! el van toeva vordben (e^{-s})

$$\tau = 1$$

$$f(t) = e^{-7t}$$

$\Downarrow$

$$\mathcal{L}^{-1}\left\{e^{-s} \frac{1}{s+7}\right\} = e^{-7(t-1)} \varepsilon(t-1)$$

$$f_3(t) = e^{-7t} \varepsilon(t) - e^{-7(t-1)} \varepsilon(t-1)$$

$$f(t-\tau) \varepsilon(t-\tau) = e^{-\tau s} F(s)$$

$$(4) F_4(s) = -\frac{e^{-2s}}{s+7} \quad (+) \text{ jeeraat}$$

$$f_4(t) = -\mathcal{L}^{-1}\left\{e^{-2s} \frac{1}{s+7}\right\} = -e^{-7(t-2)} \varepsilon(t-2)$$

eltolási
szabály
 $\tau=2$

$$e^{-7t}$$

$$(5) F_5(s) = \frac{3s^2 + 6s + 9}{s^2(s+3)}$$

$\rightarrow$ parciális törtkre bontás

$$F_5(s) = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+3} = \frac{As(s+3) + B(s+3) + Cs^2}{s^2(s+3)}$$

multiplicatív
gyök

$$s^0: 3B = 9 \rightarrow B = 3$$

$$s^1: 3A + B = 6 \rightarrow 3A + 3 = 6 \rightarrow A = 1$$

$$s^2: A + C = 3 \rightarrow 1 + C = 3 \rightarrow C = 2$$

$$F_5(s) = \frac{1}{s} + \frac{3}{s^2} + \frac{2}{s+3}$$

$$f_5(t) = \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}}_{\varepsilon(t)} + 3 \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}}_{t \cdot \varepsilon(t)} + 2 \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}}_{e^{-3t} \varepsilon(t)} = (1 + 3t + 2e^{-3t}) \varepsilon(t)$$

$$(6) F_6(s) = \frac{5s^2 + 2s + 12}{s(s^2+4)}$$

$\rightarrow$ parciális törtkre bontás

$$F_6(s) = \frac{A}{s} + \frac{Bs+C}{s^2+4} = \frac{A(s^2+4) + Bs^2 + Cs}{s(s^2+4)}$$

komplex konjugát
pár

$$s^0: 4A = 12 \rightarrow A = 3$$

$$s^1: C = 2 \rightarrow C = 2$$

$$s^2: A + B = 5$$

$$\rightarrow B = 5 - 3 = 2$$

$$F_6(s) = \frac{3}{s} + \frac{2s+2}{s^2+4} = \frac{3}{s} + 2 \frac{s}{s^2+4} + 2 \frac{1}{s^2+4}$$

$$f_6(t) = 3 \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}}_{\varepsilon(t)} + 2 \underbrace{\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\}}_{\frac{s}{s^2+\omega^2} \Big|_{\omega=2}} + 2 \mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\} =$$

$$\downarrow = \frac{1}{2} \cdot \frac{2}{s^2+4} =$$

$$\frac{\omega}{s^2+\omega^2} \Big|_{\omega=2}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+\omega^2}\right\} = \cos(\omega t) \varepsilon(t) \Big|_{\omega=2} = \cos(2t) \varepsilon(t)$$

$$\frac{1}{2} \mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\} = \frac{1}{2} \sin(\omega t) \varepsilon(t) \Big|_{\omega=2} =$$

$$= \frac{1}{2} \sin(2t) \varepsilon(t)$$

$$= (3 + 2\cos(2t) + \sin(2t)) \varepsilon(t)$$

$$(7) F_7(s) = \frac{s}{s^2+s+1} = \frac{s}{(s+1/2)^2 + 3/4} \Rightarrow \text{cos/sin atalakítás!}$$

tegyes ne'gyzet? $(s+1/2)^2 + 3/4$

$$= \frac{s + 1/2 - 1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2} = \frac{s+1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2} - \frac{1}{2} \cdot \frac{1 \cdot \sqrt{3}/2 \cdot 2/\sqrt{3}}{(s+1/2)^2 + (\sqrt{3}/2)^2} =$$

$$= \frac{s+1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2} - \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \cdot \frac{\sqrt{3}/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}$$

$1/\sqrt{3} = \sqrt{3}/3$

$$f_7(t) = \mathcal{L}^{-1}\left\{\frac{s+1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}\right\} - \sqrt{3}/3 \mathcal{L}^{-1}\left\{\frac{\sqrt{3}/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}\right\}$$

$$\frac{s}{s^2+\omega^2} \Big|_{s=s+1/2, \omega=\sqrt{3}/2} \qquad \frac{\omega}{s^2+\omega^2} \Big|_{s=s+1/2, \omega=\sqrt{3}/2}$$

$$= \left[\cos(\sqrt{3}/2 t) - \sqrt{3}/3 \sin(\sqrt{3}/2 t) \right] e^{-1/2 t} \varepsilon(t)$$

Veérték tételek

$$f(0) = \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s \cdot F(s)$$

- kezdeti érték tétele

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

- végérték / állandósult állapot tétele

de! vigyáznunk kell, hogy
van-e állandósult állapot

$$(1) F_1(s) = 3 \frac{s+2}{(s+1)(s+4)} \rightarrow f_1(t) = (e^{-t} + 2e^{-4t}) \varepsilon(t)$$

→ korábbi feladatból tudjuk

$$\begin{aligned} \text{kezdeti érték: } f_1(0) &= \lim_{s \rightarrow \infty} s \cdot 3 \frac{s+2}{(s+1)(s+4)} = \lim_{s \rightarrow \infty} 3 \frac{s^2+2s}{s^2+5s+4} = \\ &= \lim_{s \rightarrow \infty} 3 \frac{1+2/s}{1+5/s+4/s^2} = 3 \cdot 1 = \boxed{3} \end{aligned}$$

$$\text{állandósult állapot: } f_1(\infty) = \lim_{s \rightarrow 0} 3 \frac{s^2+2s}{(s+1)(s+4)} = 3 \cdot \frac{0}{4} = \boxed{0}$$

$$(2) F_2(s) = \frac{5s-12}{s^2-4s} = \frac{5s-12}{s(s-4)}$$

$$\text{kezdeti érték: } f_2(0) = \lim_{s \rightarrow \infty} s \cdot \frac{5s-12}{s(s-4)} = \lim_{s \rightarrow \infty} \frac{5s-12}{s-4} = \lim_{s \rightarrow \infty} \frac{5-12/s}{1-4/s} = \boxed{5}$$

$$\text{állandósult állapot: } f_2(\infty) = \lim_{s \rightarrow 0} \frac{5s-12}{s-4} = \frac{-12}{-4} = \boxed{3}$$

de! van állandósult állapot?

$$F_2(s) = \frac{5s-12}{s(s-4)} = \frac{A}{s} + \frac{B}{s-4}$$

$\swarrow \quad \searrow$
 $x^{-1} \quad x^{-1}$
 $A \cdot \varepsilon(t) \quad B \cdot e^{4t} \varepsilon(t) !$

eszae e^{4t} miatt

meg kell vizsgálnunk, hogy
az adott fv.-nek van-e
állandósult állapota

